

Hello, and thanks for checking out OccModWhale, our system for using occupancy modeling to track changes in humpback whale song! We hope it's as useful for your project as it has been for ours.

You can read our proposed methodology here:

<https://onlinelibrary.wiley.com/doi/10.1111/mms.70237?af=R>

Before you dive in, we recommend familiarizing yourself with the content using [occupancyTuts](#), a tutorial series about occupancy modeling and RPresence.

Things You Will Need:

- [R](#) downloaded and installed onto your machine.
- The [RPresence](#) package, which we used in R to build and run our occupancy models, downloaded and installed.
- The file FormattingData.R from this github repo.
- The file ModelingSetupLoop.R from this github repo.
- Your own song unit selection tables from Raven, Audacity, Reaper, etc. - this script uses Raven as default. See selection table formatting instructions on page 2.
- Your own recordings metadata. See recordings metadata formatting instructions on page 3.

Checklist:

- ☐ Preformat your song unit selections using FormattingData.R (instructions on page 2)
- ☐ Preformat your recordings metadata manually by setting up a descriptive excel document (instructions on page 3)
- ☐ Run ModelingSetupLoop.R
 - ☐ Decide on your number of periods
 - ☐ Decide on your number of surveys per period
- ☐ Graph your results

1. Preformatting Your Selection Tables:

To identify and distinguish between unit types, we used the system developed by Divna Djokic and Franny Oppenheimer, as described [here](#).

The formatting document FormattingData.R expects input in the form of multiple RavenPro selection tables in a single folder.

If you are using selection tables from another software (Audacity, Reaper, etc.), you will have to adjust the code accordingly.

Each selection table should have an additional annotation column in each table labeled “Designation”, into which you should have input your own unique unit label for each selection (ex. “21123”, “A”, “1.2”, “Unit1”).

If you are not using the naming conventions as detailed in Oppenheimer 2024, you may have to adjust your process accordingly (as noted in FormattingData.R).

Run FormattingData.R on your selection files. The end product should be a csv file titled “selections_metadata.csv”, with columns formatted exactly as follows if you followed the naming conventions in Oppenheimer 2024:

	Site	Begin_Time_s	End_Time_s	Contour	Tone_Type	Harmonics	Duration_s	Peak_Freq	Species
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With your data included, it should look like this:

	Site	Begin_Time_s	End_Time_s	Contour	Tone_Type	Harmonics	Duration_s	Peak_Freq	Species
1	02_Aug_2007.Table.2.selections.csv	67.06609743	67.48173115	FLAT	TONAL	DENSE	0.415633721	13791.273	31121
					...				
336	ZOOM0020_11_Aug_13.Table.2.selections.csv	28.3702057	29.90617225	FLAT	PULSED	NONE	1.535966551	3840	32321

And so on and so forth, with one row for each selection made (from all of the inputted tables combined).

2. Preformatting Your Recordings Metadata:

Your metadata for your recordings, replete with the variables you want to test through your occupancy models, should include a column for the file name, and one file for each respective variable you are testing. This script wants it as an excel sheet - the default naming convention is Recordings.xlsx.

3. Running ModelingSetupLoop.R, and occupancy modeling!

After you read in your preformatted metadata for your selections (henceforth called “detections”) and recordings (henceforth called “recordings”)

Lines 14-31: formatting for your detections dataframe. You may have to change this formatting depending on how you labeled your selections.

Lines 34-49: formatting for your recordings dataframe. You may have to change this formatting depending on the variables you are using.

Lines 51-62: assign periods to your recordings depending on the year they were made. You will have to decide how many periods you want, and how you want to split your recordings among them.

Lines 64-115: merging your dataframe (detections and recordings) into one dataframe, henceforth called “df”. This will be your main dataframe. These lines also include the production of a summary plot (“g_raw”), which details the number of detections of each species within a period.

Lines 88-10: assign surveys to your recordings depending on the number of surveys you want per period. This creates a pseudo-replication of each recording by splitting it into multiple pieces.

Lines 119-142: Creating an encounter history (“eh”) out of df.

Lines 144-180: Creating site covariates (“site_covs”) out of df.

You may have to change this code depending on the variables you want to test!

Lines 182-200: Creating survey covariates (“survey_covs”) out of recordings.

Lines 202-223: Creating a presence-absence object (“pao”) out of survey_covs

Lines 211-220: Adding in a variable regarding detection in survey2 to account for likelihood of presence being skewed in technically-related files.

Cheat Sheet for creating a Presence-Absence Object in ModelingSetupLoop.R

Input	Becomes... (1)	Becomes... (2)	Becomes... (3)	Becomes... (4)
selections_metadata.csv	detections (dataframe)	df (dataframe)	site_covs (site covariates)	pao (presence-absence object)
Recordings.xlsx	recordings (dataframe)		eh (encounter history)	pao (presence-absence object)

Input	Becomes... (1)	Becomes... (2)	Becomes... (3)
Recordings.xlsx	recordings (dataframe)	survey_covs (survey covariates)	pao (presence-absence object)

Lines 225-256: creating model sets for occupancy probability (“psimodels”) and detection probability (“pmodels”)

Lines 260-406: Occupancy modeling! Removing models with issues, creating an AIC table and a beta coefficients table.

Lines 408-647: Graphing your results!

We hope this repository is helpful for you and your project! If you have any questions or comments, feel free to reach out at franny.oppenheimer@gmail.com!